INTERNET PROGRAMMING LAB

Experiment No. 12

Title: Online Insurance Management System (InsureEasy)

AIM

To design and develop an online insurance management web application using HTML, CSS and JavaScript with a policy comparison tool, and with role-based modules: the Admin approves agents; the Agent creates customers, assigns policies and sends SMS alerts for premium dues; the Customer views policies, pays premiums, files claims and receives SMS alerts.

ALGORITHM

Step 1. Start and display the policy list with the Compare Policies tool.

Step 2. Compare: the user selects two or more policies; display a table of type, sum insured, premium, premium per Rs.1 lakh cover, term and benefits.

Step 3. Agent Registration: store the agent with status "not approved"; the agent cannot login until approved.

Step 4. Login: select role (Admin / Agent / Customer), verify credentials and open the corresponding dashboard.

Step 5. Admin: list agents with mobile, commission, number of customers and policies sold; click Approve to activate an agent login.

Step 6. Agent - Create Customer: store customer name, username and mobile linked to the agent.

Step 7. Agent - Manage Policy: select a customer and policy, enter the due date and assign the policy.

Step 8. Agent - Send SMS Alert: for each unpaid policy generate a reminder message (before or after the due date) and store it in the SMS log.

Step 9. Customer: display own policies with due date and status; Pay Premium marks the policy as paid and shows a receipt message.

Step 10. Customer: file a claim by selecting policy, amount and reason; claim status is "Submitted".

Step 11. Customer receives and views the SMS alerts sent by the agent.

Step 12. Logout and stop.

PROGRAM

File: p12.html (HTML structure, CSS styling and JavaScript logic in a single file)

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1">
<title>InsureEasy - Insurance Management</title>
<style>
:root{--c:#1565a8}
*{box-sizing:border-box;margin:0;padding:0;font-family:Segoe UI,Arial,sans-serif}
body{background:#f2f5fa;color:#222;line-height:1.4}
header{background:var(--c);color:#fff;padding:12px 20px;display:flex;justify-content:space-between;align-
items:center;flex-wrap:wrap;gap:8px}
nav button{background:none;border:0;color:#fff;margin-left:4px;padding:7px 11px;cursor:pointer;border-
radius:4px;font-size:14px}
nav button:hover{background:#ffffff33}
main{max-width:1000px;margin:18px auto;padding:0 12px}
.card{background:#fff;border-radius:8px;padding:16px;margin-bottom:16px;box-shadow:0 1px 4px #0002}
.page{display:none}.page.on{display:block}
h3{margin-bottom:10px;color:var(--c)}
input,select,textarea{padding:7px;margin:4px 6px 4px 0;border:1px solid #bbb;border-radius:4px;font-
size:14px}
.btn{background:var(--c);color:#fff;border:0;padding:8px 14px;border-radius:4px;cursor:pointer;font-
size:14px;margin:2px}
.btn.red{background:#b3261e}.btn.gray{background:#666}
table{width:100%;border-collapse:collapse;margin-top:8px;font-size:14px}
th,td{border:1px solid #ddd;padding:7px;text-align:left}th{background:#e9eef6}
.ok{color:#0a7a2f;margin:6px 0}.err{color:#b3261e;margin:6px 0}
.grid{display:grid;grid-template-columns:repeat(auto-fill,minmax(215px,1fr));gap:12px}
.item{border:1px solid #e1e4ea;border-radius:8px;padding:12px;background:#fff}
.pic{height:90px;border-radius:6px;display:flex;align-items:center;justify-content:center;font-
size:44px;background:linear-gradient(135deg,#dfe9f7,#f7e9df);margin-bottom:8px}
.pill{display:inline-block;background:#e5ecf8;border-radius:12px;padding:2px 10px;font-size:12px;margin:2px}
.stat{display:inline-block;min-width:150px;background:#eef3fb;border-radius:8px;padding:10px
14px;margin:4px;text-align:center}
.stat b{display:block;font-size:22px;color:var(--c)}
@media(max-width:600px){nav button{padding:5px 6px;font-size:12px}table{font-size:12px}}

.cmp td:first-child{font-weight:bold;background:#f3f6fb}
.sms{background:#fffbe6;border-left:4px solid #f5a623;padding:8px;margin:6px 0;font-size:14px}
</style>
</head>
<body>
<header>
  <h2>&#128737; InsureEasy</h2>
  <nav id="nav"></nav>
</header>
<main>
  <section id="home" class="page card">
    <h3>Compare Insurance Policies</h3>
    <p>Select policies to compare (Health, Auto, Home):</p>
    <div id="plist"></div>
    <button class="btn" onclick="compare()">Compare Selected</button>
    <div id="cmp"></div>
  </section>

  <section id="login" class="page card">
    <h3>Login</h3>
    <input id="lu" placeholder="Username"><input id="lp" type="password" placeholder="Password">
    <select id="lr"><option value="admin">Admin</option><option value="agent">Agent</option><option
value="user">Customer</option></select>
    <button class="btn" onclick="login()">Login</button>
    <p id="lmsg"></p>
```

```
    <small>Demo: admin/admin123  |  agent: agent1/agent123  |  customer:
cust1/cust123</small>
  </section>

  <section id="areg" class="page card">
    <h3>Agent Registration</h3>
    <input id="rn" placeholder="Agent username"><input id="rp" type="password" placeholder="Password">
    <input id="rph" placeholder="Mobile number"><input id="rc" placeholder="Commission %" type="number"
style="width:110px">
    <button class="btn" onclick="regAgent()">Register</button>
    <p id="rmsg"></p><small>Your login will work only after the admin approves it.</small>
  </section>

  <!-- ADMIN -->
  <section id="admin" class="page card">
    <h3>Admin: Approve Agent Login &amp; View Agent Details</h3>
    <div id="agtab"></div>
  </section>

  <!-- AGENT -->
  <section id="agent" class="page card">
    <h3>Agent: Create Customer</h3>
    <input id="cn" placeholder="Customer name"><input id="cu" placeholder="Login username"><input id="cph"
placeholder="Mobile">
    <button class="btn" onclick="addCust()">Create Customer</button><p id="cmsg"></p>
    <h3 style="margin-top:12px">Manage Policy Details</h3>
    <select id="pc"></select><select id="pp"></select>
    Due date <input type="date" id="pd"><button class="btn" onclick="assign()">Assign Policy</button>
    <p id="pmsg"></p>
    <div id="ctab"></div>
    <h3 style="margin-top:12px">SMS Alerts sent</h3><div id="smslog"></div>
  </section>

  <!-- CUSTOMER -->
  <section id="user" class="page card">
    <h3>My Policies</h3><div id="mytab"></div><p id="umsg"></p>
    <h3 style="margin-top:12px">File a Claim</h3>
    <select id="clp"></select><input id="cla" type="number" placeholder="Claim amount"
style="width:130px"><input id="clr" placeholder="Reason" style="width:220px">
    <button class="btn" onclick="fileClaim()">Submit Claim</button>
    <div id="claims"></div>
    <h3 style="margin-top:12px">My SMS Alerts</h3><div id="mysms"></div>
  </section>
</main>

<script>
const $ = id => document.getElementById(id);
const load = (k, d) => { try { return JSON.parse(localStorage.getItem(k)) || d; } catch (e) { return d; } };
const save = (k, v) => { try { localStorage.setItem(k, JSON.stringify(v)); } catch (e) {} };

const today = new Date(), iso = d => d.toISOString().slice(0, 10);
const plus = n => { const d = new Date(); d.setDate(d.getDate() + n); return iso(d); };
const POLICIES = [
  {id:'H1', name:'HealthPlus Gold',  type:'Health', cover:500000,  prem:9500,  term:'1 year', ben:'Cashless
hospitalisation, day-care, maternity'},
  {id:'H2', name:'FamilyShield',     type:'Health', cover:1000000, prem:16800, term:'1 year', ben:'Family
floater, pre/post hospitalisation, ambulance'},
  {id:'A1', name:'DriveSafe Comprehensive', type:'Auto', cover:800000, prem:12400, term:'1 year', ben:'Own
damage, third party, roadside assistance'},
  {id:'O1', name:'HomeSecure',       type:'Home',   cover:2500000, prem:7200,  term:'1 year', ben:'Fire,
theft, natural calamity, contents cover'}
];
let db = load('ie_db', {
  agents:[{u:'agent1', p:'agent123', ph:'98400 55555', comm:5, approved:true}],
  users:[{u:'cust1', p:'cust123', name:'Arun Kumar', ph:'98400 66666', agent:'agent1'}],
  pols:[{id:1, user:'cust1', pid:'H1', due:plus(3), paid:false}],
  claims:[], sms:[]
});
let me = null;
```

```
const pol = id => POLICIES.find(p => p.id === id);
const sv = () => save('ie_db', db);
function msg(id, t, bad) { $(id).className = bad ? 'err' : 'ok'; $(id).textContent = t; }

function nav() {
  const b = (l, p) => `<button onclick="go('${p}')">${l}</button>`;
  $('nav').innerHTML = b('Compare Policies', 'home') + (me ? b('My Dashboard', me.role) + b('Logout', 'out')
: b('Login', 'login') + b('Agent Register', 'areg'));
}
function go(p) {
  if (p === 'out') { me = null; p = 'home'; }
  document.querySelectorAll('.page').forEach(s => s.classList.remove('on')); $(p).classList.add('on');
nav();
  if (p === 'admin') renderAdmin(); if (p === 'agent') renderAgent(); if (p === 'user') renderUser();
}

/* ---------- policy comparison ---------- */
function drawList() {
  $('plist').innerHTML = POLICIES.map(p => `<label style="display:block;margin:3px 0"><input type="checkbox"
class="cp" value="${p.id}"> <b>${p.name}</b> <span class="pill">${p.type}</span> Rs.
${p.prem}/yr</label>`).join('');
}
function compare() {
  const s = [...document.querySelectorAll('.cp:checked')].map(c => pol(c.value));
  if (s.length < 2) { $('cmp').innerHTML = '<p class="err">Select at least two policies to compare.</p>';
return; }
  const row = (l, f) => `<tr><td>${l}</td>${s.map(p => `<td>${f(p)}</td>`).join('')}</tr>`;
  $('cmp').innerHTML = '<table class="cmp"><tr><th>Feature</th>' + s.map(p => `<th>${p.name}</th>`).join('')
+ '</tr>' +
    row('Type', p => p.type) + row('Sum insured (Rs.)', p => p.cover.toLocaleString('en-IN')) + row('Premium
/ year (Rs.)', p => p.prem.toLocaleString('en-IN')) +
    row('Premium per Rs.1 lakh cover', p => Math.round(p.prem / (p.cover / 100000))) + row('Term', p =>
p.term) + row('Benefits', p => p.ben) + '</table>';
}

/* ---------- login & agent registration ---------- */
function login() {
  const r = $('lr').value, u = $('lu').value.trim(), p = $('lp').value; let f = null;
  if (r === 'admin' && u === 'admin' && p === 'admin123') f = {u, role:'admin'};
  if (r === 'agent') { const a = db.agents.find(x => x.u === u && x.p === p); if (a && !a.approved) return
msg('lmsg', 'Your login is awaiting admin approval.', 1); if (a) f = {u, role:'agent'}; }
  if (r === 'user') { const c = db.users.find(x => x.u === u && x.p === p); if (c) f = {u, role:'user'}; }
  if (!f) return msg('lmsg', 'Invalid credentials.', 1);
  me = f; go(f.role);
}
function regAgent() {
  const u = $('rn').value.trim(), p = $('rp').value, ph = $('rph').value.trim();
  if (!u || p.length < 4 || !ph) return msg('rmsg', 'Fill all fields (password 4+ chars).', 1);
  if (db.agents.some(a => a.u === u)) return msg('rmsg', 'Username exists.', 1);
  db.agents.push({u, p, ph, comm:+$('rc').value || 0, approved:false}); sv(); msg('rmsg', 'Registered. Wait
for admin approval.');
}

/* ---------- admin ---------- */
function renderAdmin() {
  $('agtab').innerHTML = '<table><tr><th>Agent</th><th>Mobile</th><th>Commission</th><th>Customers</
th><th>Policies sold</th><th>Status</th></tr>' + db.agents.map(a => {
    const cs = db.users.filter(c => c.agent === a.u).map(c => c.u);
    return `<tr><td>${a.u}</td><td>${a.ph}</td><td>${a.comm}%</td><td>${cs.length}</td><td>$
{db.pols.filter(p => cs.includes(p.user)).length}</td>
    <td>${a.approved ? 'Approved' : `<button class="btn"
onclick="approve('${a.u}')">Approve</button>`}</td></tr>`;
  }).join('') + '</table>';
}
function approve(u) { db.agents.find(a => a.u === u).approved = true; sv(); renderAdmin(); }

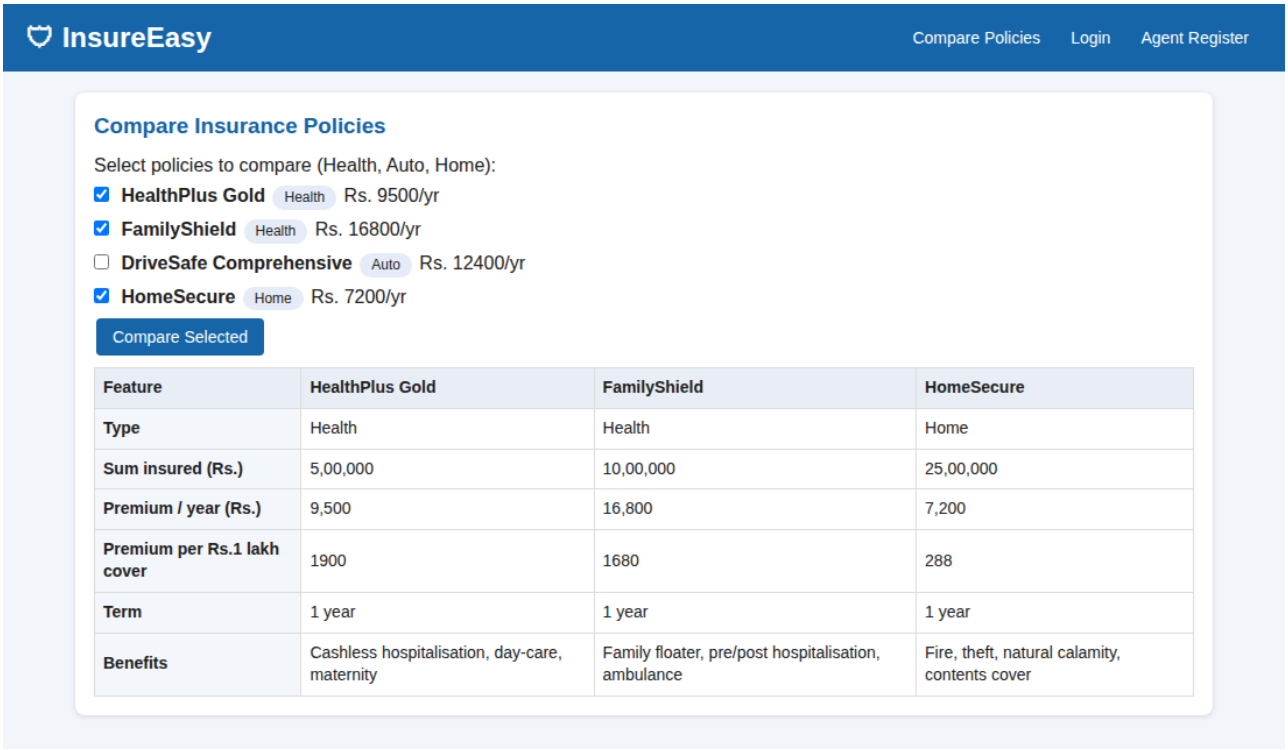
/* ---------- agent ---------- */
function renderAgent() {
  const mine = db.users.filter(c => c.agent === me.u);
```

```
  $('pc').innerHTML = mine.map(c => `<option value="${c.u}">${c.name}</option>`).join('');
  $('pp').innerHTML = POLICIES.map(p => `<option value="${p.id}">${p.name} (Rs.
${p.prem})</option>`).join('');
  $('pd').value = plus(30);
  $('ctab').innerHTML = '<table><tr><th>Customer</th><th>Mobile</th><th>Policy</th><th>Premium</th><th>Due
date</th><th>Status</th><th>Alert</th></tr>' +
    db.pols.filter(p => mine.some(c => c.u === p.user)).map(p => { const c = db.users.find(x => x.u ===
p.user), po = pol(p.pid), late = !p.paid && p.due < iso(today);
      return `<tr><td>${c.name}</td><td>${c.ph}</td><td>${po.name}</td><td>${po.prem}</td><td>${p.due}</
td><td>${p.paid ? 'Paid' : late ? '<span class="err">Overdue</span>' : 'Pending'}</td>
      <td>${p.paid ? '-' : `<button class="btn gray" onclick="sendSms(${p.id})">Send
SMS</button>`}</td></tr>`; }).join('') + '</table>';
  $('smslog').innerHTML = db.sms.filter(s => s.agent === me.u).map(s => `<div class="sms">To ${s.to}: $
{s.text}</div>`).join('') || '<p>No alerts sent.</p>';
}
function addCust() {
  const n = $('cn').value.trim(), u = $('cu').value.trim(), ph = $('cph').value.trim();
  if (!n || !u || !ph) return msg('cmsg', 'Fill all customer fields.', 1);
  if (db.users.some(x => x.u === u)) return msg('cmsg', 'Username exists.', 1);
  db.users.push({u, p:'welcome123', name:n, ph, agent:me.u}); sv(); renderAgent(); msg('cmsg', `Customer
created. Default password: welcome123`);
}
function assign() {
  if (!$('pc').value) return msg('pmsg', 'Create a customer first.', 1);
  db.pols.push({id:Date.now(), user:$('pc').value, pid:$('pp').value, due:$('pd').value, paid:false}); sv();
renderAgent(); msg('pmsg', 'Policy assigned.');
}
function sendSms(id) {
  const p = db.pols.find(x => x.id === id), c = db.users.find(x => x.u === p.user), po = pol(p.pid);
  const late = p.due < iso(today);
  db.sms.push({agent:me.u, user:c.u, to:c.ph, text:`Dear ${c.name}, your ${po.name} premium of Rs. $
{po.prem} ${late ? 'was due on' : 'is due on'} ${p.due}. Please pay to keep your policy active. -
InsureEasy`});
  sv(); renderAgent();
}

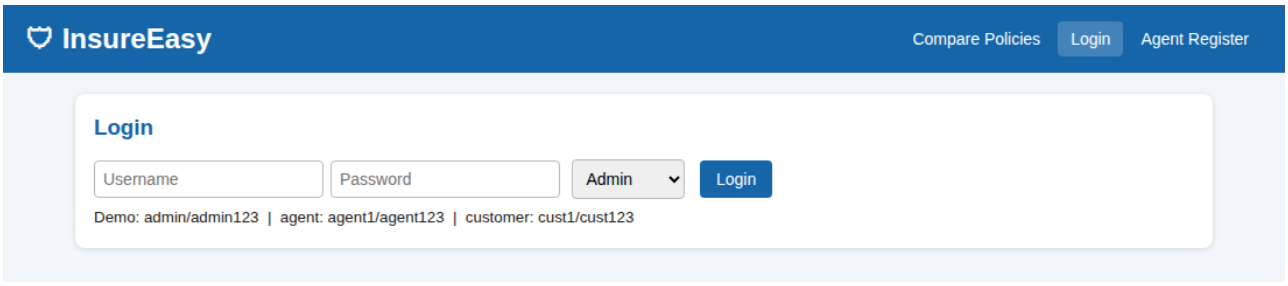
/* ---------- customer ---------- */
function renderUser() {
  const mine = db.pols.filter(p => p.user === me.u);
  $('mytab').innerHTML = mine.length ? '<table><tr><th>Policy</th><th>Type</th><th>Cover</th><th>Premium</
th><th>Due date</th><th>Status</th><th></th></tr>' + mine.map(p => { const po = pol(p.pid);
    return `<tr><td>${po.name}</td><td>${po.type}</td><td>${po.cover.toLocaleString('en-IN')}</td><td>$
{po.prem}</td><td>${p.due}</td><td>${p.paid ? 'Paid' : 'Due'}</td>
    <td>${p.paid ? '' : `<button class="btn" onclick="payPrem(${p.id})">Pay
Premium</button>`}</td></tr>`; }).join('') + '</table>' : '<p>No policies yet.</p>';
  $('clp').innerHTML = mine.map(p => `<option value="${p.id}">${pol(p.pid).name}</option>`).join('');
  $('claims').innerHTML = db.claims.filter(c => c.user === me.u).map(c => `<div class="sms">Claim #${c.id} -
Rs. ${c.amt} (${c.reason}) : <b>${c.status}</b></div>`).join('');
  $('mysms').innerHTML = db.sms.filter(s => s.user === me.u).map(s => `<div
class="sms">${s.text}</div>`).join('') || '<p>No alerts.</p>';
}
function payPrem(id) { const p = db.pols.find(x => x.id === id); p.paid = true; sv(); renderUser();
msg('umsg', 'Premium paid successfully. Receipt sent by e-mail.'); }
function fileClaim() {
  if (!$('clp').value || !(+$('cla').value > 0) || !$('clr').value.trim()) return msg('umsg', 'Choose
policy, enter amount and reason.', 1);
  db.claims.push({id:db.claims.length + 101, user:me.u, pol:$('clp').value, amt:+$('cla').value, reason:$
('clr').value, status:'Submitted'}); sv(); renderUser(); msg('umsg', 'Claim submitted.');
}

drawList(); go('home');
</script>
</body>
</html>
```

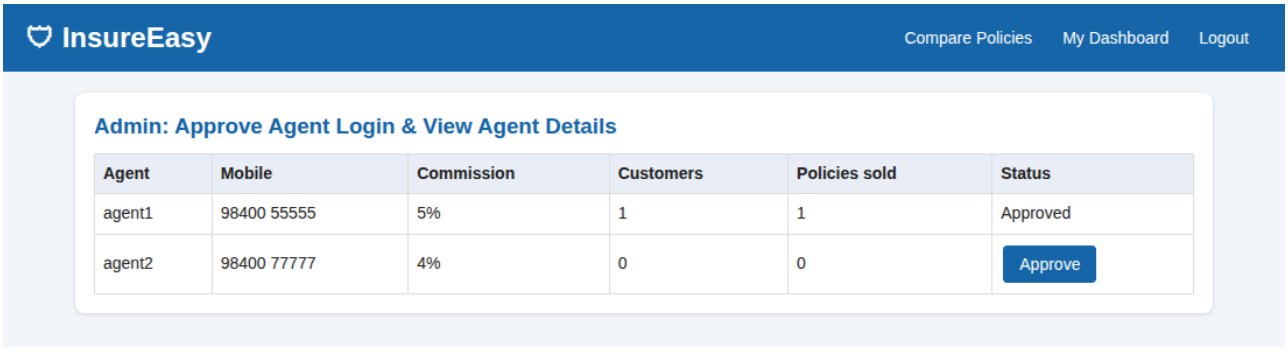
OUTPUT



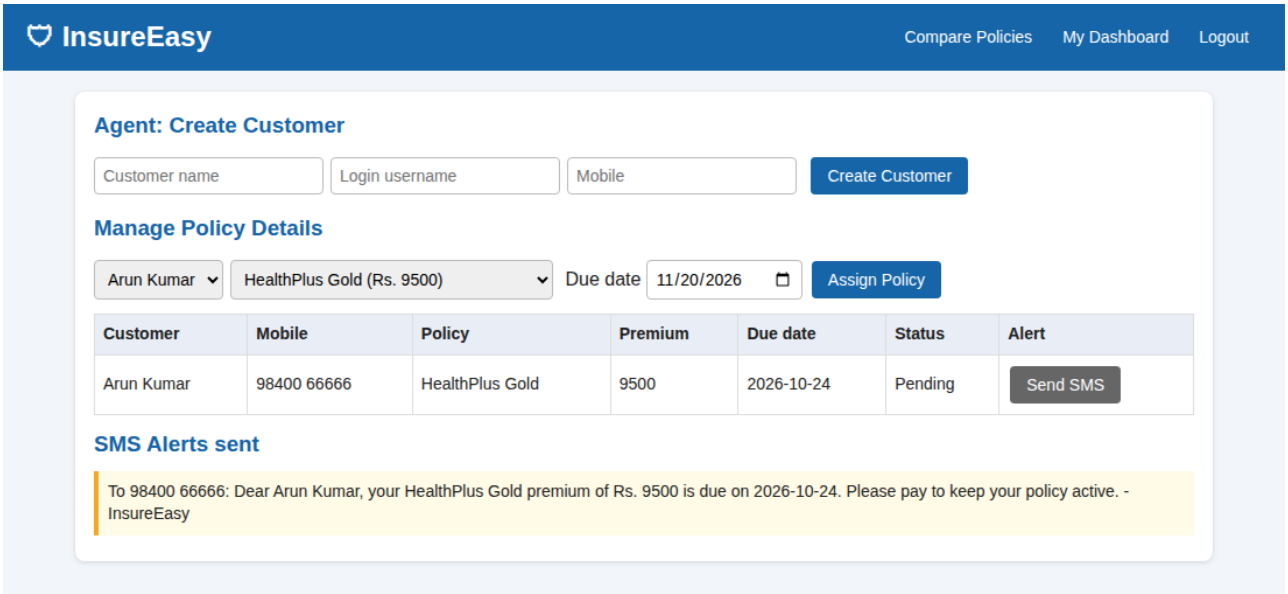
Policy comparison tool



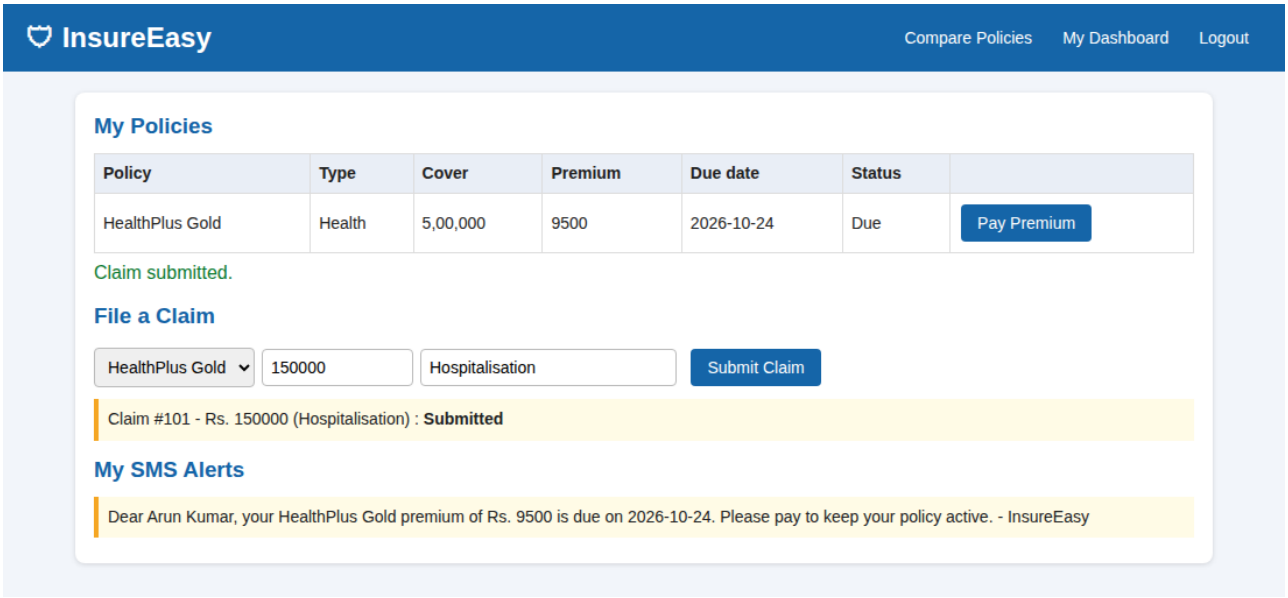
Login page with role selection



Admin - approve agent login and view agent details



Agent dashboard - customers, policies and SMS alert sent



Customer dashboard - policies, claim filing and SMS alerts

RESULT

Thus the InsureEasy insurance management website was designed and implemented using HTML, CSS and JavaScript. Policies were compared, the agent was approved, an SMS alert was sent to the policyholder and the customer filed a claim. The output was verified successfully.